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Water-energy-climate synergy: High-temperature climate resilience enhancement of mine pit pumped hydro storage through floating photovoltaics☆

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ABSTRACT

Transforming abandoned mine pits into pumped hydro storage (PHS) systems supports renewable integration while delivering economic and environmental benefits. However, high temperatures intensify PHS evaporation losses, reducing operational flexibility and grid climate resilience. Using an energy system optimization model and multi-scenario simulations, we demonstrate that deploying floating photovoltaic (FPV) systems on mine pit PHS reservoirs creates synergistic benefits under high-temperature conditions. This approach can simultaneously improve system climate resilience while controlling costs, mitigating environmental impacts, and enhancing social equity. We found that laying FPV on the surface of the "small-volume, large-area" mine pit PHS system B can fully leverage the bidirectional synergistic benefits of "water cooling-evaporation suppression": power generation efficiency is increased by up to 9.61%, and annual evaporation savings reach 27.2%. This synergistic effect significantly enhances system climate resilience, improving power supply reliability and stability by 17.69% and 24.29%. Economically, the total investment cost of the mine pit PHS system B is 37.26% lower than that of mine pit PHS system A, and FPV increases its net revenue by more than 18.4%. Furthermore, this combined system demonstrates significant comprehensive benefits in terms of land conservation, environmental protection, and social employment. Our research provides a replicable technological path for the low-carbon transformation of global fossil energy systems.

Introduction

Unchecked greenhouse gas emissions amid growing global energy demand threaten Paris Agreement climate targets [1]. As the largest high-carbon energy consumer, China must accelerate the clean energy transition by reducing coal dependence under its dual-carbon strategy [2]. Abrupt coal plant closures risk severe socio-economic consequences, including workforce displacement and regional instability [3,4]. At the same time, by innovatively transforming abandoned mine pits into pumped hydro storage (PHS) systems, the environmental risks caused by the idleness of mining land can be effectively mitigated, and the dam construction costs and land acquisition fees in the construction of PHS plants can be significantly reduced [5]. This innovative model has several successful practical cases at home and abroad [6–8]. For example, in Jurong, Jiangsu, China, the Shidangshan project repurposes an abandoned copper mine pit for PHS, cutting costs whilst saving 4,700 acres [9]. Meanwhile, the Lewis Ridge project in Kentucky, USA, is building a 237-megawatt PHS facility on former mining land. With $81 million in funding from the Department of Energy, it is expected to create 1,500 construction jobs in the area [10]. This value

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transformation from "abandoned sites" to "high-quality energy storage resources" provides a feasible path for the green transformation of traditional fossil fuel regions, offering both economic and environmental benefits [11]. However, climate change threatens system viability as extreme heat increases evaporation, diminishing storage capacity and grid resilience [12].

Hydrological processes govern reservoir evaporation, with temperature, humidity, wind speed, and solar radiation determining vapor conversion rates [13–16]. A study of 10 large reservoirs facing global water scarcity showed that some reservoirs experienced annual evaporation rates exceeding 3200 mm between 2000 and 2020 [16]. The Colorado River's Lake Mead in the southwestern United States demonstrates extreme vulnerability, losing 740 billion liters yearly to evaporation (equivalent to 300,000 Olympic pools) [17]. For PHS reservoirs with open surfaces and relatively shallow depths, the evaporation loss per unit volume is particularly significant [18]. Furthermore, the drop in water level caused by evaporation directly affects the operating efficiency of water turbines [19]. It is noteworthy that this decrease in regulatory capacity often occurs when the power grid most needs flexible resources. Extreme heat not only increases air conditioning loads and causes a surge in electricity demand but also reduces the efficiency of photovoltaic modules [20,21]. Multi-energy systems become trapped in a "peak demand–trough capacity" operational dilemma: when most dependent on PHS for grid balance, evaporation losses actually diminish its regulatory function [22]. This vulnerability in high-temperature conditions constitutes a key weakness in grid climate resilience. Building climate-resilient power systems requires deeper understanding of high-temperature impact mechanisms on PHS alongside accelerated development of evaporation suppression technologies. These parallel efforts will secure grid reliability during extreme weather events [23,24].

Floating photovoltaic (FPV) systems deployed on mine pit PHS reservoirs represent an innovative solution for high-temperature regions, delivering both evaporation control and enhanced power generation. Firstly, in water-scarce areas, FPV coverage significantly reduces evaporation losses—Egypt's Aswan High Dam achieves 49.7% evaporation reduction with 90% FPV coverage, conserving 5.9 billion cubic meters annually [25]. Secondly, the combined power generation of FPV and mine pit PHS systems creates a two-way energy efficiency improvement effect [26]. On one hand, the FPV modules benefit from the natural cooling effect of the water, effectively reducing their operating temperature and improving power generation efficiency [27–29]; on the other hand, the electricity generated by the FPV can be directly used in the pumping process, achieving on-site consumption of clean energy [30]. This synergistic model has been validated in numerous practical projects worldwide. A techno-economic assessment of three existing PHS plants conducted by Italian researchers showed that integrating an FPV system could increase the project's net present value by up to six times the benchmark level [31]. North Carolina's Tucker Town Reservoir expects to save more than 34,000 cubic meters (9 million gallons) of water yearly while generating clean electricity through its FPV installation [32]. Furthermore, FPV requires almost no additional land resources, enabling the reuse of abandoned or idle land and addressing the land constraints often faced by land photovoltaic (LPV) systems [33,34]. Therefore, deploying FPV on mine pit PHS systems in high-temperature regions is a development path that combines technical feasibility with environmental and economic benefits.

As the nation undergoing the world's largest coal power transition, China's repurposing of abandoned mines holds global significance for fossil energy decarbonization [35–37]. Developing mine pit PHS requires coordinated action: first, creating tailored deployment plans according to regional mine distribution, climate conditions, and local needs; second, deploying FPV systems on reservoirs in land-scarce, high-temperature areas to harness synergistic water-cooling and evaporation-suppression effects. However, current global research in this area is limited. This study assesses how FPV can improve the climate resilience of mine pit PHS systems under high-temperature conditions. We model two types of mine pit PHS systems combined with varying FPV coverage rates and examine their integrated impacts across technical, economic, environmental, and social dimensions. The results show that laying FPV on the surface of the "small volume, large area" mine pit PHS system B maximizes the synergistic benefits of "water cooling-evaporation suppression." This results in a maximum increase of 9.61% in FPV power generation efficiency, annual evaporation savings of 27.2%, and improvements in system power supply reliability and stability of 17.69% and 24.29%, respectively. In terms of economics, mine pit PHS system B's structural advantages deliver a 37.26% total investment cost reduction, a 27.23% levelized cost of energy storage (LCOS), and a consistent > 18.4% net revenue increase from FPV. Additional benefits include 12.76 km2 of land conservation, 651,100-tonne carbon reduction, and 3,418 new jobs. These findings demonstrate that strategic FPV integration with mine pit PHS enables simultaneous climate resilience enhancement and socioeconomic benefits, providing a replicable pathway for traditional energy regions.

This paper is structured as follows: Section 2 introduces the establishment process of the optimization model for the hybrid energy system combining the mine pit PHS system and FPV system, the selection of the study area, and the relevant parameters after scenario division. Section 3 analyzes the potential of the hybrid system in water cooling and evaporation suppression, as well as its specific applications in technical, economic, environmental, and social aspects. Section 4 summarizes the entire paper.

Methods

This section presents the detailed methodology for modelling the hybrid energy system combining mine pit PHS with FPV power generation. It encompasses the system modelling framework (Section 2.1), description of the study area (Section 2.2), and scenario design (Section 2.3). The overall modelling procedure, begins with a detailed formulation of the objective function and constraints for the energy system optimization model. Subsequently, modular modelling of each energy component is conducted, with particular emphasis on the two key characteristics of FPV systems: evaporation reduction and enhanced power generation efficiency. Finally, a multi-dimensional evaluation framework spanning technical, economic, environmental, and social aspects is established, reflecting the distinctive attributes of each power generation technology.

Hybrid energy system optimization model

Objective function

The objective of this energy system optimization and unit commitment model is to minimize the total operating cost over the optimization period. The cost function comprises various components, including fixed costs, start-up and shut-down costs, ramp-up and ramp-down costs, lost load cost, and load shedding cost. Further details are provided below:

$$\mathrm{Cost} = \min \left[\begin{array}{c} \sum_{u,i} \left(\mathrm{CostFixed}_u \times Committed_{u,i} \times TimeStep\right) + \sum_{u,i} \left(\mathrm{CostStartUpH}_{u,i} + \mathrm{Cost}ShutDownH_{u,i}\right) \\ + \sum_{u,i} \left(\mathrm{CostRampUpH}_{u,i} + \mathrm{CostRamp}DownH_{u,i}\right) + \sum_{u,i} \mathrm{CostVariable}_{u,i} \times Power_{u,i} \times TimeStep \\ + \sum_{l,i} \mathrm{Price}Transmission_{l,i} \times Flow_{l,i} \times TimeStep + \sum_{n,i} \mathrm{Cost}ShedLoad_{n,i} \times ShedLoad_{n,i} \times TimeStep \\ + \sum_{n,i} \mathrm{Cost}LostLoad \times \left(LostLoad_{n,i}^{MaxPower} + LostLoad_{n,i}^{MinPower}\right) \times TimeStep \\ + \sum_{n,i} 0.8 \times \mathrm{Cost}LostLoad \times \left(LostLoad_{n,i}^{2U} + LostLoad_{n,i}^{2D} + LostLoad_{n,i}^{3U}\right) \times TimeStep \\ + \sum_{n,i} 0.7 \times \mathrm{Cost}LostLoad \times \left(LostLoad_{n,i}^{RampUp} + LostLoad_{n,i}^{RampDown}\right) \times TimeStep \\ + \sum_{s,i} \mathrm{CostSpillage} \times \mathrm{Spillage}_{s,i} + \sum_{s,i} WaterValue \times WaterSlack_s \\ + \sum_{n,i} CurtailedPower_{n,i} \times \mathrm{Cost}Curtailment_{n,i} \times TimeStep \end{array} \right] \tag{1}$$

Where:Cost is the total operating cost of all units during the optimization period (\$);u is the generation unit in the energy system;n is the zones within each province;i is the time step in the optimization period;l is the transmission lines between nodes in the system;s is the energy storage unit that can be charged and discharged;$\mathrm{CostFixed}_u$ is the fixed cost of the generation unit in the on state (\$);$Committed_{u,i}$ is a binary variable indicating whether a generation unit is committed;$TimeStep$ is the duration of a time step of optimization;$\mathrm{CostVarieble}_{u,i}$ is the variable cost based on generation unit power output (\$/MWh);$Power_{u,i}$ is the power output generated by the generation unit (MWh);$\mathrm{CostStartUpH}_{u,i}$ is the startup cost during operation of the generation unit (\$);$\mathrm{CostShutDownH}_{u,i}$ is the shutdown cost during operation of the generation unit (\$);$\mathrm{CostRampUpH}_{u,i}$ is the cost of ramping up (\$);$\mathrm{CostRampDownH}_{u,i}$ is the cost of ramping down (\$);$\mathrm{PriceTransmission}_{l,i}$ is the unit price of transmission between different lines (\$/MWh);$Flow_{l,i}$ is the flow through lines (MWh);$\mathrm{CostShedLoad}_{n,i}$ is the penalty cost caused by the energy system adjusting the balance state of supply and demand of the system through active load shedding strategy (\$/MWh);$ShedLoad_{n,i}$ is the amount of active load shedding of the energy system (MWh);Cost$LostLoad$ is the lost load penalty cost that the energy system still causes the output to be difficult to meet the load under the condition of maximum output (\$/MWh);$LostLoad_{n,i}^{MaxPower}$ is the amount of active lost load of the energy system (MWh);$LostLoad_{n,i}^{MinPower}$ means that the output must exceed the load during some periods, mainly due to the principle of minimum total operating cost of the system and different operating characteristics of similar generator sets (MWh);$LostLoad_{n,i}^{2U}$ is the shortfall in upward reserve capacity (MWh);$LostLoad_{n,i}^{2D}$ is the shortfall in downward reserve capacity (MWh);$LostLoad_{n,i}^{3U}$ is the shortfall in non-spinning upward reserve (MWh);$LostLoad_{n,i}^{RampUp}$ is the shortfall in terms of ramping up for each unit (MWh);$LostLoad_{n,i}^{RampDown}$ is the shortfall in terms of ramping down (MWh); CostSpillage is the penalty cost of spillage from water reservoirs (\$/MWh);$Spillage_{s,i}$ is the spillage from water reservoirs (MWh); $WaterValue$ is the penalty cost that does not satisfy the storage level constraint at the end of the optimization period (\$/MWh);$WaterSlack_s$ is the unmet storage level at the end of the optimization period (MWh); $CurtailedPower_{n,i}$ is the abandoned power of renewable and non-storable energy sources such as wind and solar (MWh);Cost$Curtailment_{n,i}$ is the penalty cost of power curtailment (\$/MWh).

Constraint condition

In developing the energy system optimization model, in addition to defining the objective function, it is necessary to establish a rigorous system of constraints to ensure operational feasibility. The most fundamental and critical constraint is the real-time supply–demand balance, which requires that total power generation must dynamically match load demand and network losses during all operating periods. This constraint not only serves as a prerequisite for ensuring stable system operation but also forms the basis for all subsequent optimization calculations. Its mathematical formulation constitutes the core foundation for solving the model, as shown below:

$$\begin{aligned} &\sum_u \left(Power_{u,i} \times Location_{u,n}\right) + \sum_l \left(Flow_{l,i} \times LineNode_{l,n}\right) \\ &= Demand_{DA,n,h} + \sum_s \left(StorageInput_{s,h} \times Location_{s,n}\right) - ShedLoad_{n,i} - LostLoad_{n,i}^{MaxPower} + LostLoad_{n,i}^{MinPower} \end{aligned} \tag{2}$$

Where:DA is day-ahead;$Location_{u,n}$ is a binary variable indicating that the generation unit is in a different provincial region;$LineNode_{l,n}$ is a correlation matrix from-1 to 1, representing the power input and output of transmission lines to different provinces;$Demand_{DA,n,h}$ is the load demand of the system day-ahead market (MWh);h is the hour;$StorageInput_{s,h}$ is the charging amount per hour during the optimization period of the energy storage unit (MWh).

The minimum power output of a generation unit is typically determined by its technical minimum capacity or stable operational threshold. When the unit is committed, this constraint ensures both the stability of the generating equipment and overall system security. It also serves as a key parameter in optimization-based dispatch models.

$$PowerMin_{u,i} \times Committed_{u,i} \leqslant Power_{u,i} \tag{3}$$

Where:$PowerMin_{u,i}$ is the minimum output of the generation unit (MWh).

When a generation unit is committed, its actual power output is strictly constrained by the unit's available capacity. This technical limitation represents an important consideration in power system dispatch operations, while also forming a key boundary condition in optimization models to ensure operational feasibility.

$$Power_{u,i} \leqslant PowerCapacity_u \times AvailibilityFactor_{u,i} \times (1 - OutageFactor_{u,i}) \times Committed_{u,i} \tag{4}$$

Where:$PowerMin_{u,i}$ is the minimum output of the generation unit (MWh);$PowerCapacity_u$ is the generation unit's installed capacity (MW);$AvailabilityFactor_{u,i}$ is the percentage of available capacity of the generation unit;$OutageFactor_{u,i}$ accounts for the share of unavailable power due to planned or unplanned outages.

Other constraints for the hybrid energy system optimization model are given in Appendix 3.

Study area

This study establishes a demonstration base for a hybrid energy system integrating FPV with mine pit PHS in Tangshan City, Hebei Province. The specific geographical location and surrounding environmental characteristics are shown in Fig. S1 of supplementary information. Three primary considerations informed the site selection. First, as a traditional heavy industrial city, Tangshan possesses numerous mining-induced pits that offer natural topographic advantages for conversion into pumped storage reservoirs. Second, the region experiences a temperate monsoon climate with abundant annual solar radiation, making it suitable for large-scale photovoltaic projects. Furthermore, the well-developed grid infrastructure facilitates renewable energy integration and transmission. To accurately evaluate extreme climate impacts on system operation, we utilized the ERA5 reanalysis dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF), selecting the persistent extreme heat event during summer 2002 as a representative case. This approach provides crucial boundary conditions for investigating FPV performance under high temperatures, its water evaporation suppression effectiveness, and the flexible regulation capability of the pumped storage system. The methodology ensures both practical relevance for local energy planning and reveals operational patterns of hybrid energy systems under climate stress, thereby offering a replicable assessment framework for similar regions.

For the mine pit PHS systems, we selected two potential sites with capacities of 50 GWh and 15 GWh as our research subjects using a global map of potential PHS sites. These two sites share the same lower reservoir but possess different upper reservoirs, as illustrated in supplementary information Fig. S1 [48]. The specific parameters of the selected mine pit PHS sites are provided in supplementary information Table S1-S2.

Scenarios division

To investigate the potential of integrated FPV and mine pit PHS operation for enhancing climate resilience in hybrid energy systems, this study selects two mine pit PHS systems with distinctly different physical configurations for comparative analysis. Specifically, the upper reservoir of System A exhibits "large volume but small surface area" characteristics, while System B's upper reservoir demonstrates the opposite "small volume but large surface area" morphology. Following the principle of controlling variables, both systems maintain nearly identical geographical locations to ensure comparable meteorological conditions. The baseline hybrid system configuration comprises mine pit PHS, LPV, and wind power, representing a reference model for global high-penetration renewable energy systems. To quantitatively assess the regulation capacity of different PHS systems and reveal the "water cooling-evaporation suppression" mutual feedback mechanism between FPV and PHS, the study examines four LPV-wind capacity ratios and three FPV coverage scenarios. This multi-scenario experimental design enables systematic evaluation of synergistic effects and their impact on overall system performance. The scenario specifications are presented in supplementary information Table S3.

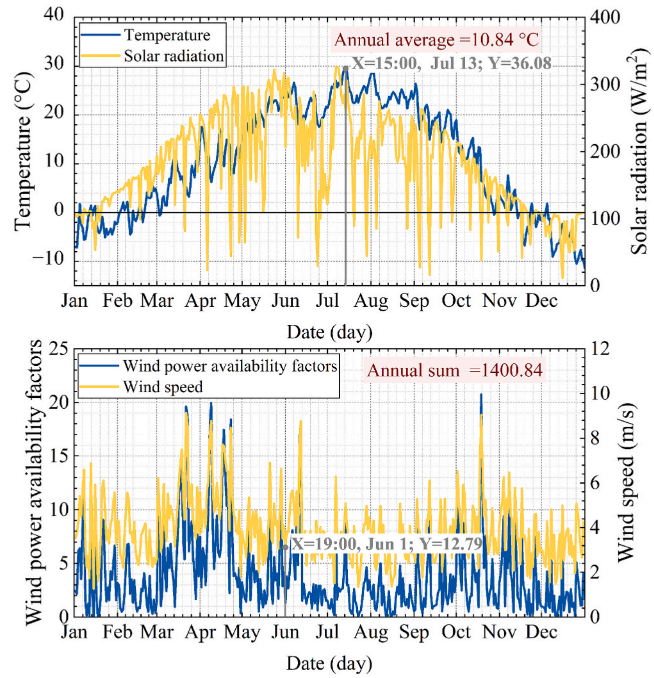

Fig. 1. Power generation potential of FPV, LPV and wind power and distribution of related climatic conditions.

Results and discussion

Power generation enhancement and evaporation reduction potential of FPV systems

In the global transition towards decarbonization, China—as a representative fossil-fuel-dependent developing economy—is actively repurposing abandoned mine pits into PHS facilities, transforming waste into valuable assets [38–40]. To reveal how FPV enhances the climate resilience of mine pit PHS under different climatic conditions, this study first compares the seasonal variations in operating temperatures and availability coefficients between LPV and FPV systems (Fig. 1, Fig. S2 of supplementary information). Results indicate that although summer brings intense solar radiation and high power generation potential for LPV systems, it also leads to elevated operating temperatures. The peak LPV operating temperature of 60.75°C was recorded at 13:00 on 14 July (Fig. S2 of supplementary information), whereas the highest ambient temperature (36.08°C) occurred earlier, at 15:00 on 13 July (Fig. 1). This difference arises because LPV operating temperature depends not only on air temperature but also on solar irradiance. Consequently, the

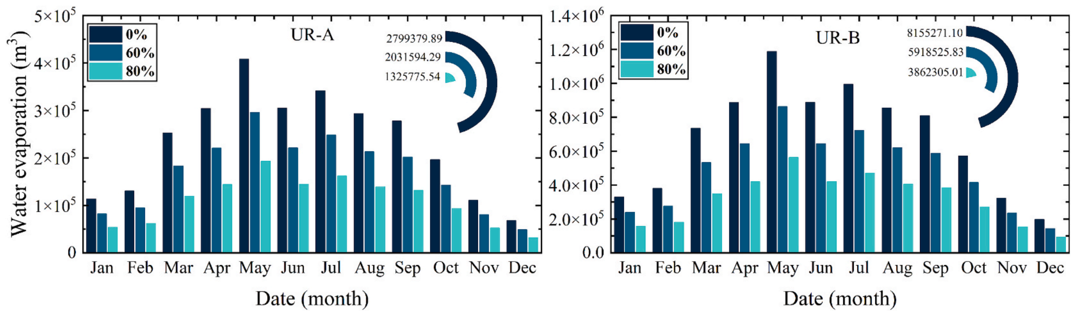

Fig. 2. Potential for evaporation reduction under different combinations of FPV coverage and mine pit PHS systems.

maximum annual availability factor of LPV (0.78) was observed at 13:00 on 24 April (Fig. S2 of supplementary information)—81 days before its peak operating temperature. These findings confirm a positive correlation between solar irradiance and LPV availability but a negative correlation with air temperature.

By contrast, the water cooling effect in PHS systems substantially lowers operating temperatures of FPV systems, boosting their power generation efficiency and availability. FPV operating temperatures respond primarily to air temperature, solar radiation, and wind speed (Fig. 1). The maximum annual FPV operating temperature reached 50.66°C at 13:00 on 30 May (Fig. S2 of supplementary information). Peak cooling performance occurred at 13:00 on 16 April, reducing FPV operating temperature by 21.36°C relative to LPV (Fig. S2 of supplementary information) while increasing power generation efficiency and availability by 9.61% (Fig. S2 of supplementary information). On a daily scale, the most effective cooling was observed on 11 June, when average FPV operating temperature dropped by 12.1°C (Fig. S2 of supplementary information) with 7.87% gains in efficiency and availability (Fig. S2 of supplementary information). Meteorological variability across temporal scales creates differing potential for efficiency gains between FPV and LPV systems. Although FPV shows particularly strong performance from April to August, its annual availability factor increases by only about 3.70% (Fig. S2 of supplementary information). Meanwhile, wind power achieves a higher annual cumulative availability factor of 1400.84 under equivalent wind speed conditions (Fig. 1).

The "water cooling-evaporation suppression" mechanism creates reciprocal benefits between FPV and mine pit PHS systems: water cooling enhances FPV efficiency whilst FPV coverage reduces reservoir evaporation, with this advantage being particularly evident under high-temperature conditions. Results show that during May, evaporation volumes from mine pit PHS system A and B without FPV coverage reached 407,816.75 m3 and 1,188,068.89 m3, respectively (Fig. 2), although peak temperatures occurred later, between June and August (Fig. 1). This is because evaporation depends not only on air temperature but also on dew point, wind speed, and both shortwave and longwave radiation. As FPV coverage increases from 60% to 100%, evaporation of mine pit PHS decreases substantially (27.43%–52.64%–100%) (Fig. 2). Notably, each 20% increase in FPV coverage nearly halves evaporation losses of mine pit PHS. The "small volume, large area" mine pit PHS system B loses 27.2% of its capacity annually without FPV coverage (Fig. 2) and would experience complete drying within three years without intervention. By comparison, mine pit PHS system A ("large volume, small area") shows only 5.40% annual evaporation without FPV, approximately one-fifth of mine pit PHS system B's rate. This demonstrates that FPV's evaporation suppression depends on both meteorological conditions and physical factors, including reservoir morphology and coverage area. For large-area reservoirs specifically, FPV's bidirectional gain substantially enhances climate resilience under

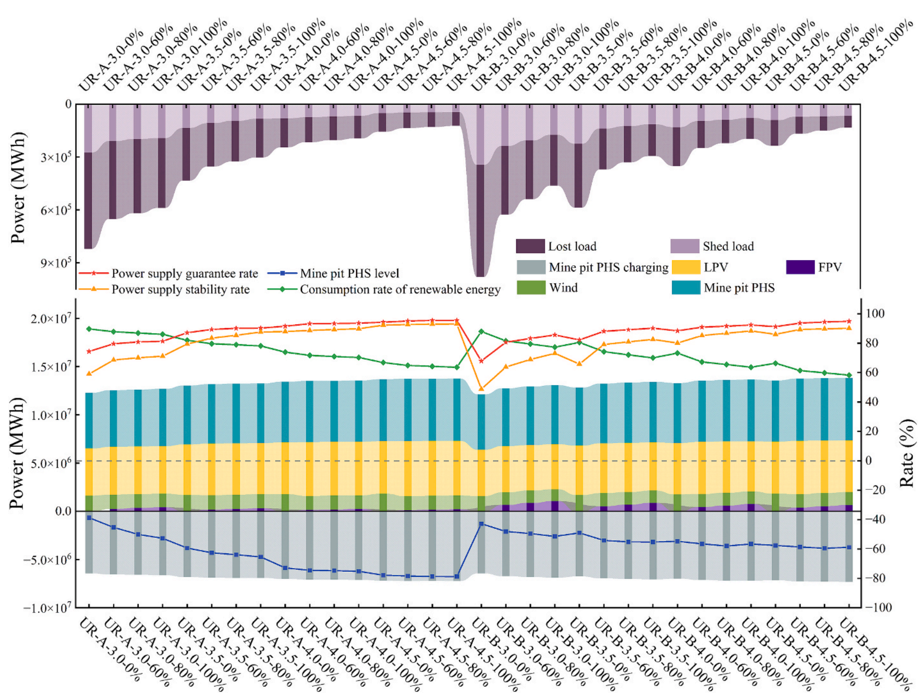

Fig. 3. FPV boosts the climate resilience potential of mine pit PHS systems in balancing power supply and demand under high temperature conditions.

extreme high temperatures. Furthermore, the different hydraulic heads (393 m for mine pit PHS system A, 204 m for mine pit PHS system B) create significant disparities in power generation from saved water. Mine pit PHS system A's annual evaporation without FPV (2,799,379.89 m3) corresponds to 2,698.13 MWh (Fig. 2, Fig. S3 of supplementary information), whereas mine pit PHS system B's evaporation volume is 2.91 times greater but produces only 1.51 times more electricity due to its lower head (Fig. S3 of supplementary information).

In summary, reservoir morphology, FPV coverage conditions, and meteorological factors, including temperature, solar radiation, and wind speed, collectively determine both the theoretical maximum of the "water cooling-evaporation suppression" synergy and the greatest climate resilience improvement achievable by FPV systems. The remaining challenge lies in optimizing FPV configuration within hybrid systems containing mine pit PHS, LPV, and wind power to fully realize its climate resilience potential.

FPV enhances the climate resilience of mine pit PHS in high-temperature climates

As renewable penetration increases in future power systems, reliable energy storage becomes crucial for maintaining supply–demand balance and enhancing climate resilience [41]. Power outages caused by generation shortfalls not only incur direct economic losses but can also trigger broader societal disruption [42]. To prevent system instability during supply deficits, proactive load-shedding strategies, including circuit breaker interventions, become necessary [43]. This study employs two core metrics—power supply guarantee rate (based on lost load duration) and power supply stability rate (based on shed load duration)—to evaluate how FPV enhances the climate resilience of mine pit PHS hybrid systems under high-temperature conditions. Results demonstrate that without FPV deployment, basic supply–demand balance can still be maintained through expanded wind and LPV capacity. When the capacity ratio between mine pit PHS system A and wind + LPV increases from 1:3.0 to 1:4.5, the system's power supply guarantee rate improves from 74.38% to 94.24%, while the stability rate rises from 59.02% to 92.23% (Fig. 3). Notably, as the wind + LPV capacity ratio increases in 0.5 increments, the growth in both reliability metrics shows diminishing returns rather than linear progression (Fig. 3). This occurs because wind and LPV generation capacity depend on meteorological conditions determining availability factors. During supply–demand imbalance periods, when availability factors remain low, even if their installed capacity is expanded, the resulting increase in power generation capacity will be relatively limited.

Compared with LPV, deploying FPV on mine pit PHS reservoirs saves land resources while also creating a "water cooling-evaporation suppression" synergy with the PHS system. Under the lowest wind power and LPV installation ratio, full FPV coverage improves the power supply guarantee rate of mine pit PHS system A by 6.91% and its power supply stability rate by 12.11% (Fig. 3). The climate resilience enhancement from FPV is particularly pronounced during high-temperature summer periods for the mine pit PHS system B hybrid system (Fig. 4, Fig. S4 of supplementary information). For the "small volume, large area" mine pit PHS system B, FPV's evaporation suppression capability and land-free deployment advantage yield increasing system value over time. It should be noted that whilst LPV capacity expansion increases renewable generation, it cannot mitigate evaporation losses from PHS reservoirs during extreme heat. Furthermore, during periods of scarce wind/solar resources and limited PHS capacity at year start/end, the hybrid system struggles to maintain supply–demand balance even with FPV deployment (Fig. 4, Fig. S4 of supplementary information).

To meet the flexibility requirements of wind power and LPV systems and to fully utilize the advantages of FPV, mine pit PHS systems must be rationally configured. This ensures a balance between supply and demand in the hybrid system while maintaining a high consumption rate of wind and solar power. Research results show that although increasing wind and solar installed capacity and adding FPV systems helps achieve supply–demand balance, it also leads to greater renewable energy curtailment. For example, the renewable energy consumption rate for mine pit PHS system A decreased continuously from 89.69% to 63.58% (Fig. 3), with curtailment mainly concentrated between March and June (Fig. S4 of supplementary information). During this period, the annual average water level in mine pit PHS system A rose rapidly from 0.39 to 0.73 (Scenario UR-A-4.0–0%), then slowed, eventually reaching 0.79 (Fig. 3), and high water levels were also focused from March to June (Fig. S4 of supplementary information). In contrast, the average annual water level of mine pit PHS system B remained relatively stable at around 0.55 (Fig. 3), with a monthly maximum of only 0.677 (Fig. S4 of supplementary information). This indicates that when wind and solar capacity is too large, mine pit PHS system A, with its substantial storage but limited regulation capability, cannot fully absorb the surplus renewable energy, leaving some of its regulation capacity idle. Notably, when the capacity of mine pit PHS system B was reduced to 30% of that of mine pit PHS system A, its minimum power supply guarantee rate was only 6.5% lower, and its load shedding rate (based on cumulative shed load) was only 1.33% higher (Fig. 3). This result further demonstrates that by scientifically configuring the mine pit PHS system, a high level of renewable energy consumption can be maintained while ensuring

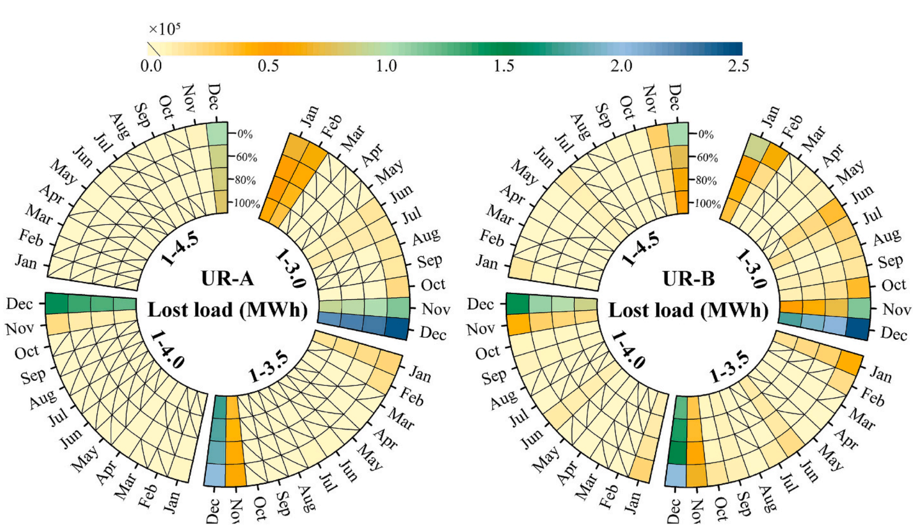

Fig. 4. The equilibrium state of supply and demand (lost load) of hybrid system under monthly scale after combined power generation by FPV and mine pit PHS system.

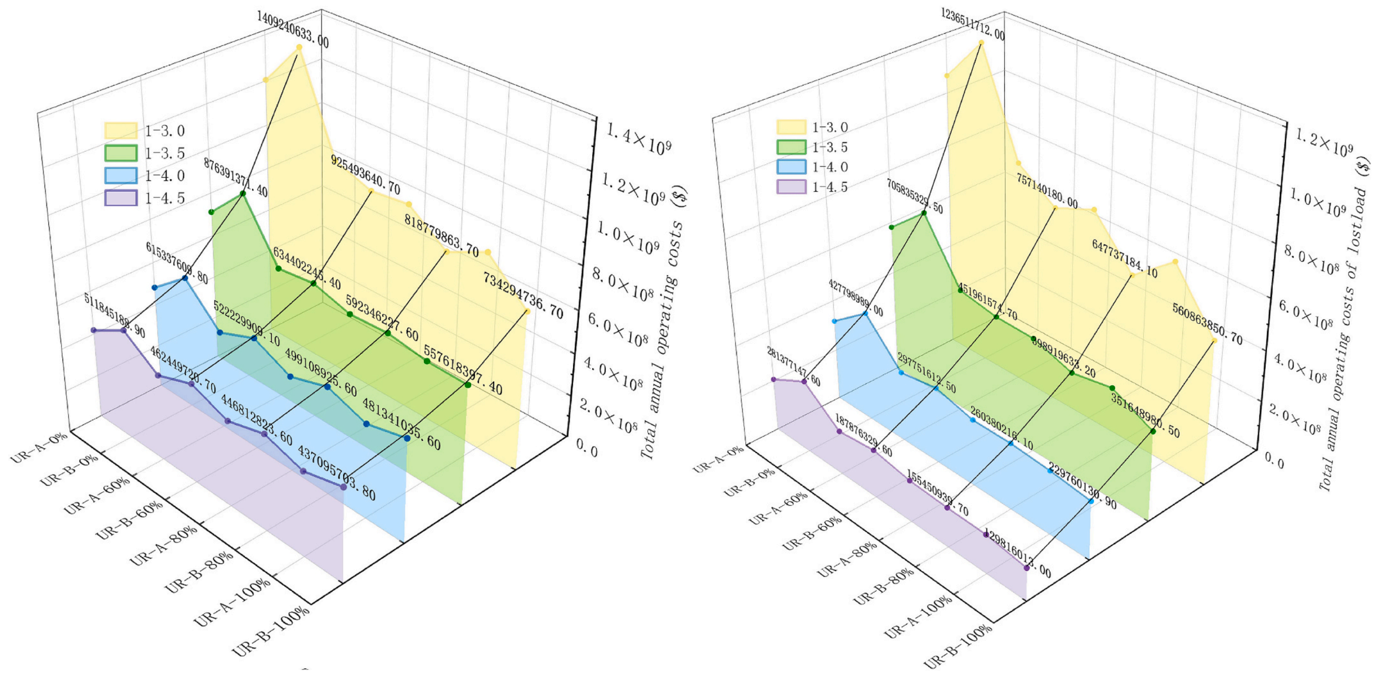

Fig. 5. Economic characteristics of hybrid systems in terms of total operating costs at annual scales.

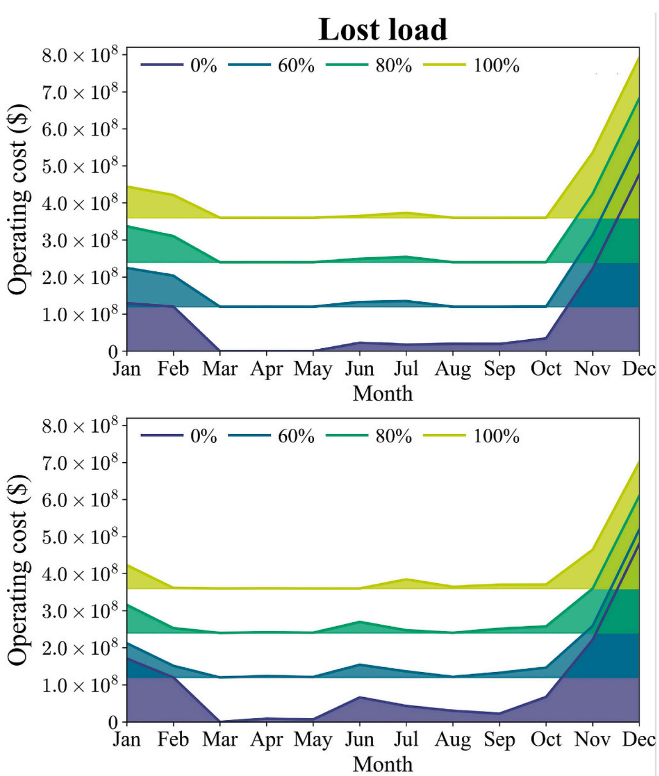

Fig. 6. Monthly allocation of operating costs related to lost load in hybrid systems.

supply–demand balance.

Impact of FPV coverage on hybrid system operating costs

The above research shows that expanding the installed capacity of wind power, LPV, and FPV can effectively enhance the climate change adaptation capacity of hybrid systems with mine pit PHS. To determine the optimal output scheme under different scenarios, this study uses total operating cost as the objective function and systematically analyzes the impact pathway of adding FPV on operating costs. Based on the weight analysis of various cost parameters and their unit prices, the three parameters with the greatest impact on total operating cost are identified as lost load cost, shed load cost, and curtailment cost.

Results show that under the combined effect of various cost parameters, the hybrid system with mine pit PHS system B, at the minimum wind and solar installed capacity and without FPV coverage, had an annual total operating cost of $1.409 billion (Fig. 5) and a net profit of $-720 million (Fig. 7). This outcome occurred primarily as lost load cost accounted for 87.74% of total operating cost (Fig. 5). High lost load cost led to continuous losses when the total installed capacity ratio of wind and LPV was only 3/4 (Fig. 7). As wind, solar, and FPV installed capacity expanded, the total operating cost significantly decreased due to a substantial reduction in lost load, with a maximum decrease of 69.89% (Fig. 5). However, the decreasing trend gradually slowed, consistent with changes in the three major cost parameters (Fig. 3). After covering the surface of mine pit PHS system B with FPV, the most direct effect was that the system's net profit, originally $-720 million, was largely eliminated (Fig. 7), benefiting from its extensive water surface area and the advantages of FPV not occupying land and being compatible with LPV. This advantage is particularly prominent in summer. For example, when the total installed capacity ratio of wind and LPV was 3/4, full FPV coverage completely eliminated the lost load cost in June ($66.055 million) and reduced the shed load cost in August ($9.086 million) by 87.71% (Fig. 6, Fig. S6 of supplementary information). Furthermore, as the total installed capacity ratio of wind and LPV increased to 4.5/5.5, the weight of annual curtailment cost in the total operating cost began to exceed that of lost load cost (Fig. 5, Fig. S5 of supplementary information). On a monthly scale, even at the lowest wind and solar installed capacity ratio, systems with mine pit PHS system A and B exhibited this phenomenon in March–May and August–October for mine pit PHS system A and March–May for mine pit PHS system B, respectively (Fig. 6, Fig. S6 of supplementary information). This indicates that the impact of wind and solar installed capacity on lost load and curtailment is mainly reflected on an annual scale, while on a monthly scale, they show a complementary relationship. Notably, without FPV coverage, the annual curtailment cost proportion of mine pit PHS system A was lower than

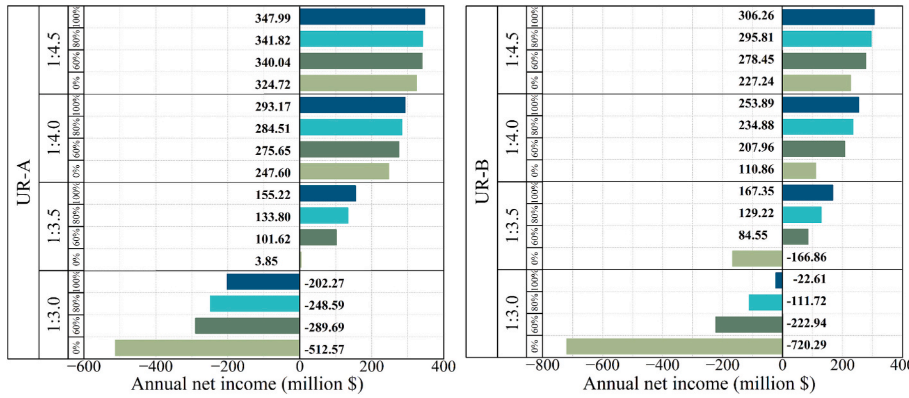

Fig. 7. Comprehensive net benefits of hybrid systems after considering power generation benefits and operating costs.

that of the smaller-capacity mine pit PHS system B, with a maximum difference of 7.9% (Fig. S5 of supplementary information). Although the unit price of shed load cost is high, because the total shed load does not exceed 5% of the total load, its annual cost proportion reached a maximum of only 9.26% (Fig. S5 of supplementary information). It can be seen that blindly expanding wind and solar installed capacity not only has a limited effect on improving supply–demand balance but also leads to substantial curtailment losses. This is because, on one hand, the PHS regulation capacity is insufficient to absorb the excess wind and solar power; on the other hand, when the storage capacity is too large, the PHS releases water to generate electricity to utilize curtailed power, which instead occupies the space for wind and solar power generation, leading to new rounds of curtailment.

To quantify the economic performance of the two mine pit PHS systems, this paper introduces investment cost and the LCOS as core evaluation indicators. Mine pit PHS system A, with its larger storage capacity and head, incurs higher energy storage cost ($898 million) and tunnel cost ($1.642 billion). Combined with plant costs ($1.041 billion), its total investment reaches $3.583 billion. In contrast, the corresponding costs for mine pit PHS system B are $230 million, $572 million, and $1.445 billion, respectively, with a total investment of $2.248 billion. The lower investment cost gives mine pit PHS system B a much lower LCOS of just $23.82 per MWh, compared to $32.73 per MWh for system A. Additionally, the "small volume, large area" characteristic of mine pit PHS system B makes the improvement in system net revenue more significant after FPV coverage. For instance, when the total installed capacity ratio of wind and LPV is 4/5, the economic gain from FPV for mine pit PHS system A falls below 15% (Fig. 7), while for mine pit PHS system B it remains no less than 18.4% (Fig. 7).

Overall, mine pit PHS system A possesses greater storage capacity but lower utilization, and as wind and solar capacity expands, growing idle capacity leads to poorer economic performance. In contrast, mine pit PHS system B achieves better economic returns owing to its larger FPV coverage area and lower investment cost. In the long term, the evaporation suppression effect of FPV will further enhance these advantages. This demonstrates that through rational configuration of mine pit PHS systems with appropriate FPV coverage, significant economic benefits can be achieved while improving climate resilience under high-temperature conditions.

Environmental and social outcomes of integrated energy systems

Beyond established technical and economic benefits, FPV systems also offer significant environmental and social advantages [44]. To systematically quantify these contributions, we evaluate environmental indicators—including land saving, direct and indirect water saving, evaporation reduction ratio, carbon emission reduction, and coal saving—alongside the social indicator of employment creation. The most notable environmental advantage is FPV's minimal land use, which enables PV capacity expansion at higher generation efficiency without competing with LPV installation. This is particularly valuable when land around PHS sites is unsuitable for LPV, avoiding remote siting that hinders grid integration—a constraint effectively mitigated by FPV. Our results show that covering 60%, 80%, and 100% of mine pit PHS system A's surface with FPV saves 2.63 km2, 3.50 km2, and 4.38 km2 of land, respectively (Fig. 8). The savings are substantially greater for mine pit PHS system B, at 7.66 km2, 10.21 km2, and 12.76 km2—approximately three times those of mine pit PHS system A (Fig. 8).

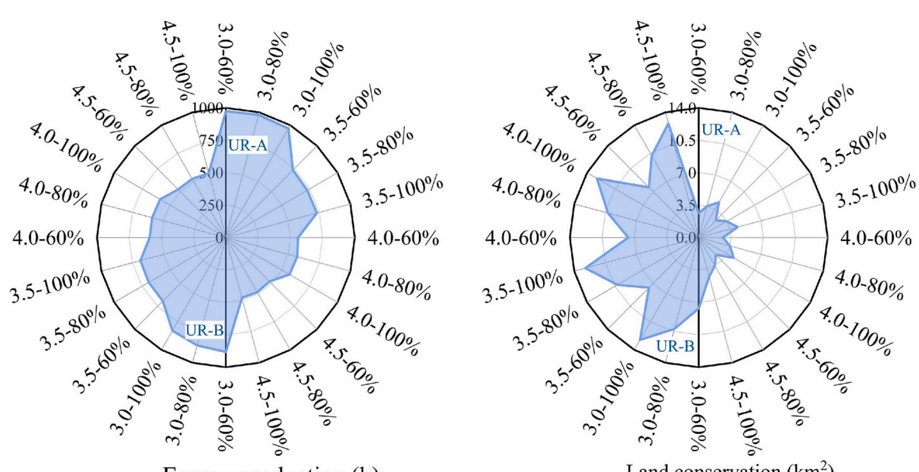

Fig. 8. After the mine pit PHS system covers FPV, the system has comprehensive benefits in energy production and land conservation.

Another key environmental characteristic is FPV's evaporation suppression effect, influenced by water area, reservoir morphology, and local climate. Research indicates the minimum annual evaporation saving for mine pit PHS system A is 0.7678 million m3 (Fig. S7 of supplementary information), representing 1.48% of its total storage capacity (Fig. S7 of supplementary information), with a maximum saving of 2.7994 million m3 (5.40%). Mine pit PHS system B demonstrates more substantial savings, with minimum and maximum annual evaporation reductions of 2.2367 million m3 (7.46% of capacity) and 8.1553 million m3 (27.20%), respectively. As FPV coverage increases, the evaporation inhibition rate relative to the equivalent open water area reaches 45.71%, 65.80%, and 100% (Fig. S7 of supplementary information), while the proportion of total water area covered reaches 27.43%, 52.64%, and 100% (Fig. S7 of supplementary information). Unlike direct water saving, FPV's indirect water saving capacity depends on actual power generation. The study finds maximum indirect water savings for mine pit PHS systems A and B under the "3.0–100%" scenario are 296 million m3 and 1.417 billion m3, respectively (Fig. S7 of supplementary information). This difference occurs because the model allocates FPV generation according to installed capacity share, which decreases as wind and LPV capacity increases. Related environmental indicators follow similar trends: maximum carbon emissions for mine pit PHS systems A and B are 19,000 and 47,300 tonnes, respectively (Fig. S7 of supplementary information), while maximum carbon emission reductions reach 261,600 and 651,100 tonnes (Fig. S7 of supplementary information), and maximum coal savings are 122,800 and 305,800 tonnes (Fig. S7 of supplementary information). Finally, FPV deployment generates social benefits—full coverage of mine pit PHS systems A and B can create 1,173 and 3,418 employment opportunities, respectively.

Our comprehensive analysis shows that the integrated FPV and mine pit PHS system delivers multiple advantages. For resource utilization, FPV's minimal land footprint effectively reduces land pressure, while its evaporation suppression significantly improves water conservation. Regarding environmental benefits, both direct and indirect water savings, carbon emission reductions, and coal savings all increase with system scale. Furthermore, project construction and operation create substantial employment opportunities. Overall, the FPV and mine pit PHS hybrid system achieve synergies across energy, resource, and social domains, offering an integrated regional sustainability solution.

Conclusions

This study develops an optimization model for hybrid energy systems incorporating two mine pit PHS configurations with varying FPV coverage levels. It first reveals how FPV enhances power generation efficiency while suppressing water evaporation across different PHS configurations. The research then evaluates FPV's role in strengthening climate resilience and identifies key pathways affecting economic performance in hybrid systems. Finally, it systematically quantifies FPV's comprehensive environmental and social benefits, including water conservation, carbon reduction, coal savings, and employment generation. The main findings are as follows:

(1) The mutually beneficial "water cooling-evaporation suppression" relationship between FPV and mine pit PHS is driven by reservoir morphology and meteorological conditions. During April-August high-temperature periods, water cooling is most effective, reducing FPV operating temperature by up to 21.36°C while improving generation efficiency and availability factor by 9.61%. Meanwhile, evaporation suppression is particularly effective in the "small volume, large area" mine pit PHS system B, achieving maximum annual evaporation savings of 27.2%.

(2) Although FPV installation enhances climate resilience without additional land use, it also increases curtailment risk. While full FPV coverage on mine pit PHS system A maximizes improvements in power supply guarantee rate (6.91%) and stability rate (12.11%), the system still struggles to maintain supply–demand balance during winter when wind and solar resources are scarce. During summer, mine pit PHS system A's limited regulation capability leaves partial capacity idle, further exacerbating curtailment.

(3) Rational configuration of mine pit PHS with appropriate FPV coverage maintains good economic performance while enhancing high-temperature climate resilience. Mine pit PHS system B achieves a levelized cost of storage of $23.82/MWh, benefiting from lower energy storage ($230 million) and tunnel costs ($572 million). Furthermore, FPV consistently improves net revenue by over 18.4% for this system, highlighting its synergistic advantages.

(4) The integrated FPV and mine pit PHS system demonstrates multidimensional benefits across resource utilization, environmental protection, and social development. For mine pit PHS system B, the system saves 12.76 km^2 of land, achieves annual direct water savings of 8.155 million m^3, indirect water savings of 1.417 billion m^3, carbon emission reductions of 651,100 tonnes, and creates 3,418 jobs, demonstrating comprehensive sustainable development value.

While this study systematically analyzes and quantifies the multidimensional impacts of integrated FPV and mine pit PHS operation across technical, economic, environmental, and social aspects, certain limitations should be noted. First, although the research focuses on system-level operational benefits, the potential effects of FPV coverage on reservoir microclimate and aquatic ecology have not been examined. Therefore, when promoting the application of this scheme, it is recommended to conduct long-term ecological monitoring and environmental assessment in conjunction with specific water areas in order to comprehensively balance energy efficiency and ecological sustainability. This is also a key area that future research needs to focus on. Second, the social benefit assessment mainly addresses employment creation, without considering psychosocial factors such as community acceptance and visual landscape impacts, which may influence public support and long-term viability in practical implementation. This study primarily focuses on the synergistic benefits of mine pit PHS systems and FPV under high-temperature conditions. Its applicability to other climatic conditions (such as cold, rainy regions) or different mine morphologies (such as deep, narrow mines) has not yet been explored. Future research could extend the analytical framework to diverse climatic scenarios and mine types to assess its universality and explore system design strategies under multi-objective optimization.

CRediT authorship contribution statement

Yuanqiang Gao: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Jinhui Zhang:** Writing – review & editing, Resources. **Haozhe Xu:** Writing – review & editing, Resources. **Yongjie Liu:** Writing – review & editing, Resources. **Wenxin Xie:** Writing – review & editing, Resources. **Diyi Chen:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendices

1. Calculation of parameters related to the FPV system.

1. Installed capacity of FPV.

The installed capacity of FPV systems is calculated using a standardized energy density parameter of 99.23 MWp·km^{-2} [45]. The calculation process primarily considers two core parameters: first, the available water surface area determined through field surveys, and second, the energy density coefficient, which comprehensively incorporates technical characteristics including photovoltaic panel efficiency, installation tilt angle, and array spacing.

$Capacity_{FPV} = 99.23 \times Area_{FPV}$(1)where: $Capacity_{FPV}$ is the installed capacity of FPV (MW);$Area_{FPV}$ is the area covered by the FPV on the surface of the mine pit PHS water domain (km^2).

2. The potential of FPV to improve power generation efficiency.

The water-cooling effect resulting from the combined operation of FPV systems and mine pit PHS is quantified using an improved empirical model [46]. This model incorporates both heat exchange mechanisms and hydrodynamic characteristics, significantly enhancing temperature prediction accuracy. It thereby provides an important basis for system performance evaluation.

$$T_{FPV} = 2.0458 + 0.9458 \times T_a + 0.0215 \times Sr - 1.2376 \times Ws \tag{2}$$

where: T_{FPV} is the operating temperature of FPV (℃);T_a is ambient temperature (℃); Sr is solar radiation (W·m^{-2}); Ws is the wind speed (m·s^{-1}).

For comparative purposes, this study employs the empirical relationship proposed in Eq. (13) to characterize the correlation between operating temperature and ambient air temperature in LPV systems [47]. This model provides an important benchmark for evaluating the performance of conventional photovoltaic systems.

$$T_{LPV} = T_a + \frac{(T_{NOCT} - 20)}{800} \times Sr \tag{3}$$

where:T_{NOCT} is the operating temperature of the LPV under the test operating conditions, which is set to 44 ℃ here.

Building upon the identified cooling effects, a quantitative relationship between the power generation efficiency of FPV and LPV systems can be established. The corresponding mathematical formulation is presented below:

$$\eta_{FPV} = \eta_{LPV}[1 - \alpha_P(T_{FPV} - T_{LPV})] \tag{4}$$

where:η_{FPV} is the power generation efficiency of FPV operation process (%);η_{LPV} is the power generation efficiency of LPV operation process (%);α_P is the temperature coefficient that quantifies how photovoltaic module efficiency changes with operating temperature, and is assigned a value of 0.0045 in this study.

3. Evaporation reduction potential of FPV.

The shading effect from FPV systems effectively reduces reservoir evaporation losses. This study applies the Penman-Monteith equation (shown below) to calculate natural evaporation rates. By integrating energy balance and mass transfer principles, this approach provides a reliable method for quantifying the actual water conservation benefits of FPV installations.

$$ET = K_\omega \frac{0.408\Delta(R_n - G) + \gamma \frac{37}{T_a+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \tag{5}$$

where:ET is the natural evaporation of the open water when the mine pit PHS does not cover the FPV (mm·h^{-1});K_ω is the open water coefficient, set to 1.05;Δ is the slope of saturated water vapor pressure curve (kPa·℃$^{-1}$);R_n is the net radiation at the water surface (MJ·m^{-2}·h^{-1});G is the water heat flux density (MJ·m^{-2}·h^{-1}).γ is the psychometric constant (kPa·℃$^{-1}$);e_a is the actual vapor pressure (kPa) and e_s is the saturated vapor pressure (kPa);.u_2 is the average hourly wind speed (m·s^{-1}).

2. Comprehensive evaluation system.

1. Economics of mine pit PHS.

To systematically evaluate the economic performance of energy storage technologies, we employ the LCOS as our primary metric. This indicator calculates the cost per unit of discharged energy by distributing total lifecycle costs, incorporating key parameters such as initial capital expenditure, annual operation and maintenance expenses, energy losses, and system lifespan. The specific calculation formula is presented below [48]:

$$LCOS = \frac{I + \sum_{t=0}^{n}(M_t + L_t)/(1 + r)^t}{\sum_{t=1}^{n} E_t/(1 + r)^t} \tag{6}$$

where:I is the initial investment cost of the mine pit PHS plant ($);$M_t$ is the maintenance cost incurred during the operation of the mine pit PHS plant ($); L_t is the energy loss cost caused by different charging and discharging efficiencies ($);$E_t$ is the energy generated by mine pit PHS plant generation (MWh);r is the discount rate (%);t is the life span;n is the expected lifespan.

The economic assessment of power plants is primarily based on key engineering parameters, which form the foundation for investment cost calculations. The available reservoir volume depends on two core factors: the system's energy storage requirement and the design head. This relationship shows that for a given energy storage capacity, the effective reservoir volume is inversely proportional to the head parameter, as quantified in the following formula [48]:

$$V_{\text{usable}} = \frac{E \times 3600}{9.81 \times \eta_{\text{gen}} \times H} \tag{7}$$

where: V_{usable} is available storage capacity (million m^3);E is the storage capacity (GWh);H is the head (m); η_{gen} is the generation efficiency, set to 0.9.

The total investment cost of a mine pit PHS plant comprises several key components. It primarily consists of energy storage costs (covering reservoir infrastructure and electromechanical equipment installation), tunnel construction costs (involving water conveyance systems, pressure pipelines and support structures), powerhouse construction costs (including underground cavern excavation and surface power generation buildings), land acquisition and resettlement costs, as well as initial water filling costs (covering required water resources and water treatment facilities for initial impoundment). Together, these cost elements form the fundamental framework for project economic evaluation [48].

$$V_{\text{total}} = \frac{V_{\text{usable}}}{\eta_{usable}} \tag{8}$$

$$V_{rock} = \frac{V_{\text{usable}}}{R_{water}} \tag{9}$$

$$Area_{reservoir} = \frac{200 \times V_{total}}{D_{\text{average}}} \tag{10}$$

$$\text{Cost}_{reservoir} = \frac{V_{rock} \times C_{dam}}{\text{f}_{\text{inflation}}} \tag{11}$$

$$\text{Cost}_{\text{fill}} = \frac{V_{\text{total}} \times Price_{\text{initial}}}{1000} \tag{12}$$

$$\text{Cost}_{land} = \frac{\text{Area}_{reservoir} \times \text{Price}_{land}}{100} \tag{13}$$

$$\text{Cost}_{tunnel} = \frac{(66000P + 170000000) + 1000D \times (1280P + 210000) \times H^{-0.54}}{1000000 f_{\text{inflation}}} \tag{14}$$

$$\text{Cost}_{\text{powerhouse}} = \frac{63.5 \times H^{-0.5} \times P^{0.75}}{\text{f}_{\text{inflation}}} \tag{15}$$

$$Cost_{total} = Cost_{\text{reservoir}} + Cost_{\text{fill}} + Cost_{\text{land}} + Cost_{\text{tunnel}} + Cost_{\text{powerhouse}} \tag{16}$$

where: V_{usable} is available storage capacity ($10^6 m^3$);V_{total} is total storage capacity ($10^6 m^3$);η_{usable} is the usable fraction of the water volume, set to 0.9; V_{rock} is the earthwork volume required by the energy storage power plant ($10^6 m^3$);R_{water} is the ratio of combined water to rock; $Area_{\text{reservoir}}$ is the mixed area of the energy storage power plant (Ha); $D_{average}$ is the average reservoir depth (m); $\text{Cost}_{\text{reservoir}}$ is the energy storage costs of the reservoir ($\$10^6$); C_{dam} is the unit cost of building a dam ($\$/m^3$), set to 168;$\text{f}_{inflation}$ is the inflation factor, set to 0.85; $\text{Cost}_{\text{fill}}$ is the earthwork cost ($\$10^6$); $\text{Price}_{initial}$ is the price of initial fill acquisition ($/ml), set to 25; $\text{Cost}_{\text{land}}$ is the land acquisition cost ($\$10^6$); Price_{land} is the unit price of land acquisition ($\$/m^2$), set to 0.1;$\text{Cost}_{\text{tunnel}}$ is the cost of water conveyance tunnels ($\$10^6$);$P$ is the installed capacity (MW);D is the distance between the closest points of the upper and lower reservoirs (m);$\text{Cost}_{\text{powerhouse}}$ is the cost of power plant buildings ($\$10^6$);$Cost_{total}$ is the total investment cost ($\$10^6$).

2. Evaluation index of FPV.

This study establishes a multi-indicator quantitative framework covering environmental and social dimensions to systematically evaluate the comprehensive benefits of FPV systems. The assessment focuses on four key metrics: water conservation (including direct evaporation suppression and indirect water savings), carbon reduction potential, displacement of conventional coal-fired power generation, and employment creation opportunities. This multi-dimensional assessment framework enables a comprehensive understanding of the synergistic benefits of FPV in advancing energy transition and sustainable development [49].

$$EY = \frac{E_{annual}}{\text{Capacity}_{FPV}} \tag{17}$$

$$CE = E_{annual} \times k_{LCE} \tag{18}$$

$$CS = E_{annual} \times k_{CS} \tag{19}$$

$$JC = \text{Capacity}_{FPV} \times k_{job} \tag{20}$$

$$CR = E_{annual} \times k_{CR} \tag{21}$$

$$WS_{indirect} = \frac{0.75 \times 3600 \times E_{annual} \times 10^6}{\rho \times g \times H} \tag{22}$$

where: EY is the energy production (h);E_{annual} is the annual electricity generation of FPV (MWh); CE is the carbon emission of the FPV (t); k_{LCE} is the life-cycle rate of solar PV system, set to 0.045; CS is the coal saving of FPV (t); k_{CS} is the coal utilization coefficient of thermal power plants (t/MWh), set to 0.291; JC is the number of jobs created by FPV;k_{job} is the job creation coefficient (jobs/MW), set to 2.7; CR is the carbon reduction potential of FPV (t);k_{CR} is the carbon reduction potential coefficient (t/MWh), set to 0.6198; $WS_{indirect}$ is the indirect water saving of FPV (m^3).

3. The other constraint condition of the hybrid energy system optimization model.

In addition to the constraints applied to conventional generation units, energy storage systems must also satisfy specific operational requirements. The first constraint stipulates that the energy content of a given storage unit must remain above a defined minimum threshold. This condition prevents excessive discharge, ensures operational stability, and represents a fundamental boundary condition in optimization models.

$$StorageMin_s \leqslant StorageLevel_{s,i} \times Nunits_s \tag{23}$$

where:$StorageMin_s$ is the minimum storage level of the energy storage unit (MWh);$StorageLevel_{s,i}$ is the storage level during the operation of the energy storage unit; $Nunits_s$ is the number of units

For energy storage units, the availability factor applies to both charging/discharging power and storage capacity. This technical parameter directly influences the system's actual dispatch potential and operational reliability, while also serving as an important parameter for building accurate optimization models:

$$StorageLevel_{s,i} \leqslant StorageCapacity_s \times AvailabilityFactor_{s,i} \times Nunits_s \tag{24}$$

where:$StorageCapacity_s$ is the storage capacity of the energy storage unit (MWh).

The energy added to storage units is constrained by their charging capacity. Charging is only permitted while the units are not simultaneously generating power, ensuring operational safety and optimizing equipment utilization.

$$StorageInput_{s,i} \leqslant \mathrm{StorageChargingCapacity}_s \times \left(Nunits_s - Committed_{s,i}\right) \tag{25}$$

where:$\mathrm{StorageChargingCapacity}_s$ is the installed charging capacity of the energy storage unit (MW).

Power flow between nodes is strictly limited by transmission line capacity. In optimization dispatch models, this constraint serves as a critical condition for ensuring solution feasibility, directly related to maintaining thermal stability limits and ensuring secure system operation. Furthermore, this limitation significantly influences both congestion management and market clearing outcomes in power systems.

$$FlowMin_{l,i} \leqslant Flow_{l,i} \leqslant FlowMax_{l,i} \tag{26}$$

where:$FlowMin_{l,i}$ is the minimum flow in line (MWh);$FlowMax_{l,i}$ is the maximum flow in line (MWh).

Data availability

Data will be made available on request.

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